

56. GLUCOSE-FRUCTOSE RATIO: CHECK WHAT YOU BUY!

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Knowledgeiswatt

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In the previous KisW research notes, we saw that:

- Carbohydrates ingestion during exercise improves performance in long races (generally above 2 hrs).
- The responsible physiological mechanism seems to be glycogen (endogenous carbs, which is a very limited resource in the body) sparing,

Read more [here](#) about the above points.

However, we also saw that there is no linear dose-response for the amount of carbs ingested and performance. So, no higher the carbs ingested, better the performance (read more [here](#)).

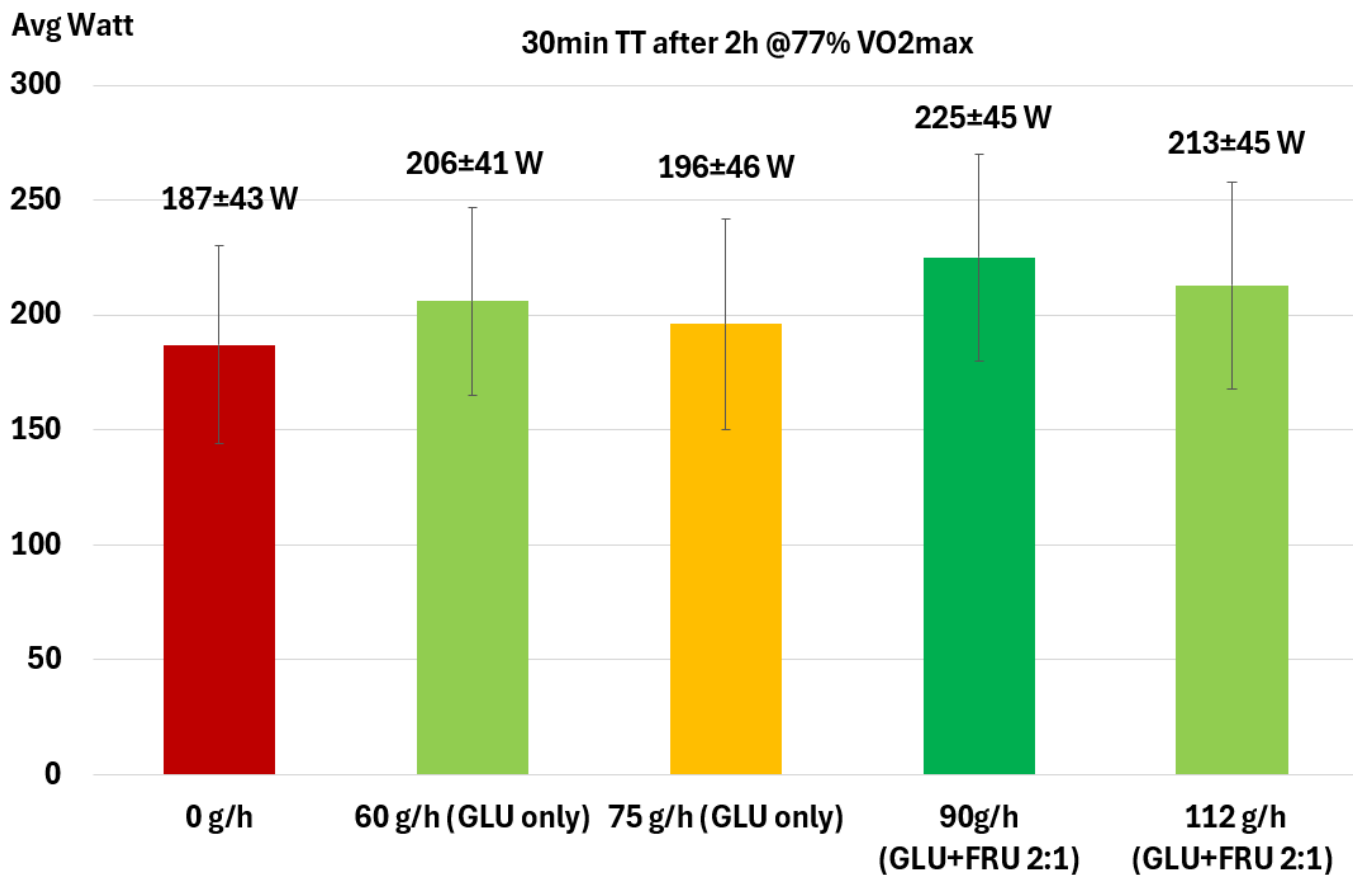
Why?

While some factors like body size, diet and gut training can influence this number (1), **glucose gut transporters get generally saturated at 60 grams per hour (g/h)**. When going over this threshold, there is an increase in the risk of gastrointestinal distress without any additional positive benefit on performance.

The good news is that

fructose use a different gut transporter than glucose, so when going over 60 g/h you should add fructose.

This should be the reason behind the performance data on amateur cyclists reported by King et al. 2018 (2). They showed the differences in avg power during a 30 min TT after riding 2hrs at 77% of VO2max, while ingesting different amounts and types of carbohydrates during exercise:



Data from King et al. 2018.

We can see that:

- Performance with 75 g/h Glucose only vs 60 g/h Glucose only is not increased. This is likely because we are overdosing glucose gut transporter.
- Performance in 90 g/h with glucose to fructose ratio 2:1 (60g glucose + 30g fructose) is increased compared to 60 g/h GLU only. This is likely because we add 30 g of fructose than can be used in top of what should be glucose transporter maximum capacity (60 g/h).
- Performance in 112 g/h with glucose to fructose ratio 2:1 (75g glucose +37 g fructose) is not further increased compared to 90 g/h with the same glu-fru ratio. This is because they probably overdosed the glucose transporter while using not so different fructose amounts: 30 vs 37g.

So, what is the science-based practical tip about the carbs to ingest during prolonged exercise to optimize performance?

My suggestion is to check the carbohydrates composition of the bar, gel or powder you are going to buy or to bring with yourself during the ride, using these guidelines:

- **When you target 60 g/h or lower:**

you can just don't care about the type of carbs you ingest (glucose vs fructose). You should be able to absorb everything. Note that this threshold value (60 g/h) could depend also on your body size (read more [here](#)).

- **Between 60 and 90 g/h:**

glucose to fructose ratio of 2:1 should be used. In this way you are adding fructose while being sure to not exceed 60 grams per hour of glucose.

- **When going over 90 g/h:**

You should increase the ratio of fructose ingested, so a glucose to fructose ratio of 1:0.8 or even 1:1 (if total carbs amount is higher than 120 g/h) should be used to not overdose glucose transporters.

Thanks for reading! Spread with your team!

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REFERENCES:

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1. King AJ, O'Hara JP, Morrison DJ, Preston T, King RFGJ. Carbohydrate dose influences liver and muscle glycogen oxidation and performance during prolonged exercise. *Physiol Rep*. 2018 Jan;6(1):e13555.
 2. Ijaz A, Collins AJ, Moreno-Cabañas A, Bradshaw L, Hutchins K, Betts JA, Podlogar T, Wallis GA, Gonzalez JT. Exogenous Glucose Oxidation During Exercise Is Positively Related to Body Size. *Int J Sport Nutr Exerc Metab*. 2024 Sep 27:1-12.



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